



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Control Statements in Programming Languages



Learning Outcome:

Exchange information about logic, its principles, and methods in simple spoken and written situations.

Describe the development of software programming in simple spoken and written situations.

Ask your partner:

- Why do you think each programming language uses different codes to achieve certain task?
- Why do you think control statements are important in programming?

Can Do Statements:

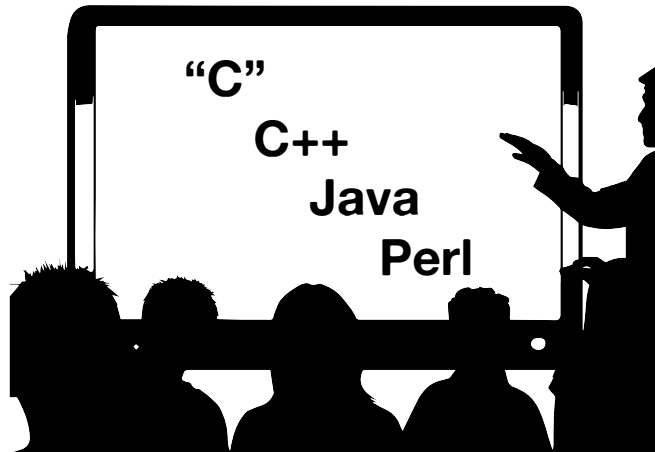
- I can list the types of control statements.
- I can say how to structure conditional statements in different programming languages.
- I can discuss basic information about iterative loops.





Let's Explore!

This class is about control statements.



Instructions: Read the text below.

Teacher: Today we are talking about some programming concepts. Let's start with control statements. Why are they important? Well, you can't execute the sections of code in a program without using control statements. They are in charge of controlling the execution of those sections of code you want. Before using them, it is important to understand the types of control statements. Can you tell me something about them?

Kate: There are three: Sequential, decision and iterative. Sequential, which is the default control structure, is the way in which you execute the statements one after the other in a certain order. Decision, or conditional - referred to as if, then and else - depends on the performance of a logical test that leads the statement to handle certain programming situations. With decision statements you can't get a result without alternative execution of instructions. Finally, the iterative is the one in charge of controlling the number of times a code is executed.

Teacher: Great, Kate! I imagine you can also tell me how to structure conditional statements in different programming languages...

Kate: Well, I know you can't do it without boolean expressions...

Roger: If I'm not mistaken, all programming languages use specific statements. The condition particle is a boolean expression that will evaluate if certain computation is true or false. You can't have only one possibility! There are many ways to structure it according to the program's configuration.

Kate: Teacher, I want to know more about iterative constructs or loops. I know they control and iterate the number of repetitions a code is executed.

Teacher: Correct. They are divided in FOR loops and WHILE/DO loops. When you write FOR loops you need three factors: the initialization, the condition and the increment. WHILE/DO loops only repeat a section of code if the conditions is true; if it is not, it ends





Let's Practice!

I know there are three types of control statements but...

Instructions: Complete the sentences by writing its corresponding word. Use: can't understand, can't include, can't listen, can't eat.

1. I have to do a report about the conditional statements, but I _____ all of them.
2. We know the sequential is the default control structure, but I _____ the decision type.
3. They talk about the iterative, but I _____, I am far from them.



Let's Practice!

There are many ways to structure programming languages.

Instructions: Select the correct answer to complete the sentences.

1. We can't _____ what iterative constructs or loops are about.
 - a. remembering
 - b. remember
 - c. to remember
2. Iterative constructs -or loops- control and _____ many times the number of sequences in which a code is executed.
 - a. delete
 - b. repeat
 - c. can't
3. You can classify into 2 categories: FOR loops and WHILE/DO loops. They can't _____ more.
 - a. be
 - b. are
 - c. is
4. If you write FOR loops, you need three factors: the initialization, the condition, and the increment. You can't _____ them.
 - a. forget
 - b. wait
 - c. eat



Extension Activity

Conditional Statements in Different Programming Languages

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about structure conditional statements in different programming languages.
- Identify different features of these programming languages.
- Write down a paragraph in which you include the most important characteristics of these programming languages.
- Share the paragraph with your instructor.

Self-Assessment

Yes No

- I look for information about structure conditional statements in different programming languages.
- I identify different features of programming languages.
- I write down a paragraph including important characteristics.
- I share the paragraph with my instructor.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **computation** (noun): the result of a computing method
e.g. The condition particle is a Boolean expression that will evaluate if certain computation is true or false.
- **Iterate** (verb): to repeat something several times
e.g. I know they control and iterate the number of repetitions a code is executed.
- **If I'm not mistaken** (expression): used to indicate you are not totally certain about something.
e.g. If I'm not mistaken, all programming languages use specific statements.
- **execution** (noun): the act of accomplishing something
e.g. Control statements are in charge of controlling the execution of those sections of code you want.
- **performance** (noun): the doing of a duty
e.g. It depends on the performance of a logical test that leads the statement to handle certain programming situations.
- **lead** (verb): to cause
e.g. The logical test leads the statement to handle certain programming situations.



Learning Points

Verb Can (Permission, Possibility, Ability) in Negative Statements

Can is a modal verb that expresses permission, possibility or ability in the present.

Use **can** when you want to ask for a permission.

For example:

Can I ask a question?

You also use **can** to talk about possibilities.

For example:

I **can** be there at 6 am.

Finally, use **can** to express ability.

For example:

She **can** remember every conditional code in Java programming language.

In order to make the negative form, you just have to use **cannot** or its contraction form **can't**.

For example:

She **can't** use Boolean operators. (She has not the ability to use Boolean operators)

Examples

- *You can't execute the sections of code in a program without using control statements.*
- *With decision statements, you can't get a result without alternative execution of instructions.*



Learning Points

Conditional Codes in Different Programs

Control statements use certain codes to express a conditional statement according to the different programming language.

The following chart shows the conditional codes that the different programming languages use.

Programming Language	Conditional Code
Pascal	if (<i>condition</i>) then
	begin
	<i>statements</i>
	end;
"C", C++, Java, Perl	if (<i>condition</i>) {
	<i>statements</i>
	} else {
	<i>else_statements</i>
Oracle PL/SQL	if (<i>condition</i>) then
	<i>statements</i>
	else
	<i>statements</i>
Basic	If <i>condition</i> Then
	<i>statements</i>
	Else
	<i>statements</i>
	End If